

Last Time:

- Rotating Frames
- Rotation Matrix Kinematics
- Velocity Transport Theorem
- Frictional Forces
- MEMS Gyro

Today:

- Finish MEMS Gyro
- Rigid Bodies
- Inertia Matrix
- Kinetic Energy

MEMS Gyro:

- Full Dynamics from last time:

$$m\ddot{\mathbf{r}} + \underbrace{2m\omega \times \dot{\mathbf{r}}}_{\text{Coriolis}} + \underbrace{m\ddot{\omega} \times \mathbf{r}}_{\text{Transverse}} + \underbrace{m\omega \times \omega \times \mathbf{r}}_{\text{Centrifugal}} + \underbrace{k\mathbf{r}}_{\text{Spring}} + \underbrace{c\dot{\mathbf{r}}}_{\text{damper}} = \underbrace{\begin{bmatrix} a \sin(\omega_0 t) \\ 0 \\ 0 \end{bmatrix}}_{\text{drive}}$$

* Some Simplifications

1) Assume $\omega_z \ll \omega_0 = \sqrt{\frac{k}{m}}$ ($\omega_0 \sim 20 \text{ KHz}$ for MEMS gyros)

2) Assume $\dot{\omega}_z$ small

3) Ignore z-axis dynamics

$$\Rightarrow \begin{bmatrix} \ddot{r}_x \\ \ddot{r}_y \end{bmatrix} + 2 \begin{bmatrix} 0 & -\omega_z \\ \omega_z & 0 \end{bmatrix} \begin{bmatrix} \dot{r}_x \\ \dot{r}_y \end{bmatrix} + \frac{k}{m} \begin{bmatrix} r_x \\ r_y \end{bmatrix} + \frac{c}{m} \begin{bmatrix} \dot{r}_x \\ \dot{r}_y \end{bmatrix} = \begin{bmatrix} \frac{a}{m} \sin(\omega_0 t) \\ 0 \end{bmatrix}$$

$$\Rightarrow \ddot{r}_x + \frac{\kappa}{m} r_x + \frac{c}{m} \dot{r}_x = \frac{a}{m} \sin(\omega_0 t) + 2\omega_z \dot{r}_y$$

small compared to drive

$$\underbrace{\ddot{r}_y + \frac{\kappa}{m} r_y + \frac{c}{m} \dot{r}_y}_{\text{damped SHO}} = \underbrace{-2\omega_z \dot{r}_x}_{\text{drive proportional to } \omega_z}$$

$\Rightarrow$ Amplitude of r_y is proportional to ω_z

- We can measure r_y and (knowing $\ddot{r}_x$) calculate ω_z

Rigid Bodies

- A collection of point masses whose relative distances are fixed:

$$\|\vec{r}_i(t) - \vec{r}_j(t)\| = \underbrace{C_{ij}}_{\text{constant}} \quad \forall t, \quad i, j \in [1 \dots N]$$

- Never true in reality

- Practical engineering definition: Frequencies associated with flexible modes $\gg$ frequencies associated with rigid-body motion (e.g. controller bandwidth)

- Often situation dependent:

1) Launch vehicles typically have lowest bending mode around 1-10 Hz. Trajectory planner can treat this as a rigid body, but tracking controller must be filtered/band limited to avoid exciting flexible modes.

2) Car on highway $\Rightarrow$ rigid body vs. race car where suspension dynamics matter.

- * Rigid body models are extremely useful across aerospace and robotics (aircraft, underwater vehicles, rockets, spacecraft).

Transformations + Configuration / Pose

- A particle only has position, but a rigid body has more configuration/pose information.
- We could keep track of N particle positions (and $\sim O(N^2)$ constraints), but that's inefficient.
- What are all the possible transformations I can do to the N particles that will respect the relative-distance constraints?

* Translation:

$$r_i' = r_i + d \quad \forall i, \quad d \in \mathbb{R}^3$$

$$\|r_i' - r_j'\| = \|r_i + \cancel{d} - r_j - \cancel{d}\| = \|r_i - r_j\| = C_{ij}$$

* Rotation:

$$r_i' = Q r_i, \quad Q \text{ is a rotation matrix } (Q \in SO(3))$$

$$\|r_i' - r_j'\| = \|Q r_i - Q r_j\| = \|Q(r_i - r_j)\|$$

$$= \sqrt{(r_i - r_j)^T \underbrace{Q^T Q}_I (r_i - r_j)} = \sqrt{(r_i - r_j)^T (r_i - r_j)} = \|r_i - r_j\| = C_{ij}$$

$\Rightarrow$ We can just keep track of a single translation + rotation instead of N positions

- Translation + rotation called "rigid-body transformation"

Kinetic Energy

- For a particle, $T = \frac{1}{2} \dot{\tilde{\mathbf{r}}}^T \dot{\tilde{\mathbf{r}}}$

- If we sum over all particles in a rigid body:

$$T = \sum_{k=1}^N \frac{1}{2} m_k \dot{\tilde{\mathbf{r}}}_k^T \dot{\tilde{\mathbf{r}}}_k$$

- Since all particles must translate + rotate together:

$$\tilde{\mathbf{r}}_k(t) = \tilde{\mathbf{Q}}(t)^B \mathbf{r}_k + \tilde{\mathbf{d}}(t)$$

$\tilde{\mathbf{Q}}$ constant

$$\dot{\tilde{\mathbf{r}}}_k(t) = \dot{\tilde{\mathbf{Q}}}^B \mathbf{r}_k + \dot{\tilde{\mathbf{d}}} = Q S(\omega)^B \tilde{\mathbf{r}} + \tilde{\mathbf{v}}$$

- Plug this into T :

$$\begin{aligned} T &= \sum_{k=1}^N \frac{1}{2} m_k (\dot{\tilde{\mathbf{r}}}_k)^T (\dot{\tilde{\mathbf{r}}}_k) = \sum_{k=1}^N \frac{1}{2} m_k (Q S(\omega)^B \tilde{\mathbf{r}} + \tilde{\mathbf{v}})^T (Q S(\omega)^B \tilde{\mathbf{r}} + \tilde{\mathbf{v}}) \\ &= \underbrace{\sum_{k=1}^N \frac{1}{2} m_k \tilde{\mathbf{v}}^T \tilde{\mathbf{v}}}_{(1)} + \underbrace{\sum_{k=1}^N m_k \tilde{\mathbf{v}}^T Q S(\omega)^B \tilde{\mathbf{r}}}_{(2)} + \underbrace{\sum_{k=1}^N \frac{1}{2} m_k \tilde{\mathbf{r}}^T S(\omega)^T Q^T (\dots)}_{(3)} \end{aligned}$$

1) This is just kinetic energy of a particle with mass:

$$M = \sum_{k=1}^N m_k$$

2) I can re-arrange the sum slightly:

$$\tilde{V}^T Q S(\omega) \left[\sum_{k=1}^N m_k {}^0r_k \right]$$

definition of center of mass in the body frame

⇒ If I chose origin of B frame to be at CoM, this term is zero. We (almost) always do this by convention.

3) Let's re-arrange to move ${}^B\omega$ outside the sum:

$$\sum_{k=1}^N \frac{1}{2} m_k r_k^T S(\omega) \underbrace{Q^T Q}_I S(\omega) {}^0r_k$$

$= -S({}^B r_k) \omega$

$$= \frac{1}{2} \omega^T \left[\sum_{k=1}^N m_k \underbrace{S({}^0r_k)^T S({}^0r_k)}_{\text{"inertia Matrix" } J} \right] \omega = \frac{1}{2} \omega^T \left[-\sum_{k=1}^N m_k S({}^B r_k) \right] \omega$$

$$= \frac{1}{2} {}^0\omega^T J {}^B\omega$$

$$\Rightarrow \boxed{T = \frac{1}{2} M \tilde{V}^T \tilde{V} + \frac{1}{2} {}^0\omega^T J {}^B\omega}$$

- Note that J depends on choice of B frame axes (just like ${}^B\omega$).